

Measuring the radius of a star — Solution

Y23 (2023) — English translation

A1

0.20 pts

Let us consider the case when rays 1 and 2 are parallel to the z axis, and mirror M3 is parallel to the xy plane. Find the difference in optical paths Δ_{12} between rays 1 and 2 if they both fall on the screen at point O .

In horizontal sections, the path difference is $L_1 - L_2$, and the first ray also travels a distance L_3 twice. Thus, the total difference is:

$$\Delta_{12} = L_1 + 2L_3 - L_2$$

Answer:

$$\Delta_{12} = L_1 + 2L_3 - L_2$$

A2

1.40 pts

Let us consider the case when the mirror M3 is deflected by an angle α from the plane xy by rotating around the axis x (see Fig. 1). If ray 1' hits the screen at point O , find the optical path difference $\Delta_{1'2}$ between rays 1' and 2. Show that, to a first approximation, the path difference is independent of α and simplify your answer at $\alpha \ll 1$.

Note that the triangle obtained from the paths of rays 1 and 1' from the mirror M3 to the point O is isosceles. Also, changes in the horizontal and vertical paths due to changes in the point of incidence of the beam compensate each other. The only parameter affecting the path traveled by beam 1 is the coordinate z of the point of impact on the mirror M3: it affects the distance traveled from HM to M3. But the optical path of beam 2 does not change in any way. We get:

$$\Delta_{1'2} = \Delta_{12} - (2(L_3 + L_4) \sin \alpha) \cdot \sin \alpha = \Delta_{12} - 2(L_3 + L_4) \sin^2 \alpha$$

Substitute the result of the previous part:

$$\Delta_{12'} = L_1 + 2L_3 - L_2 - 2(L_3 + L_4) \sin^2 \alpha \approx L_1 + 2L_3 - L_2 - 2(L_3 + L_4) \alpha^2$$

Answer:

$$\Delta_{1'2} = L_1 + 2L_3 - L_2 - (L_3 + L_4) \frac{\sin \alpha \sin 2\alpha}{\cos \alpha} = L_1 + 2L_3 - L_2 - 2(L_3 + L_4) \alpha^2$$

B1

0.80 pts

Let rays 1 and 2 fall at fixed points of mirrors M1 and M2, respectively. Show in the solution sheets that, to a first approximation, the optical paths of rays 1 and 2 from the points of reflection from the mirrors M1 and M2, respectively, to the screen do not depend on the angles θ and α .

B2

0.80 pts

Let the angles α and θ be fixed, and the positions of the points of incidence of rays 1 and 2 on the mirrors M1 and M2, respectively, can change. Show in the solution sheets that the optical paths of rays 1 and 2 from the dotted line corresponding to the coordinate $z_d = L_3 + L_4$ (meaning the points of intersection of the dotted line with the trajectories of rays 1 and 2 before falling on the mirrors M1 and M2, respectively) to the screen, to a first approximation, do not depend on the angles θ and α .

B3

1.00 pts

Express b and δ_O in terms of λ , L_1 , L_2 , L_3 , L_4 .

δ_O can be found using the previous part of the problem, since this is the value of the phase difference for $\theta = 0$. Note that the phase difference arises not only due to the difference in optical paths, but also due to reflections: beam 1 is reflected one time more, so it is additionally shifted by π .

$$\delta_O = \frac{2\pi}{\lambda}(L_1 + 2L_3 - L_2) + \pi$$

One can note that, due to the statements proved in B1 and B2, for the change of angle θ difference phase arise only from due to change path to intersection with dashed line $z_d = L_3 + L_4$, that is:

$$\frac{2\pi}{\lambda}(L_1 + L_2)\theta = \frac{2\pi}{\lambda}b\theta$$

$$b = L_1 + L_2$$

Answer:

$$b = L_1 + L_2 \delta_O = \frac{2\pi}{\lambda}[L_1 + 2L_3 - L_2] + \pi$$

B4

0.60 pts

Show in the solution sheets that

$$\delta_y = \delta_O - \frac{2\pi}{\lambda} \cdot 2\alpha y$$

Note: Further, in all points of the problem, you can use the statements of points B1-B4, even if they are not proven.

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Consider two rays 1 that arrive at points on the screen located at a distance y . Due to the smallness of α , the distance between the points of reflection of these rays from M3 is also equal to y . Note that the difference in optical paths between them arises only because of the different coordinates z of the reflection points from M3: the path from HM to M3 and from M3 to the screen is shortened.

$$\Delta L = -2y\alpha$$

$$\delta_y = \delta_O - \frac{2\pi}{\lambda} \cdot 2\alpha y$$

Answer:

$$\delta_y = \frac{2\pi}{\lambda}[L_1 + 2L_3 - L_2 - 2\alpha y] + \pi$$

B5

0.60 pts

Thus, an interference pattern will appear on the screen. Find the distance y_1 to the interference maximum closest to point O . Also find the distance Δy_m between two adjacent interference maxima. Give numerical answers in the case of $L_1 = 0.50$ m, $L_2 = 1.50$ m, $L_3 = 0.50$ m, $L_4 = 1.00$ m, $\alpha = 10^{-4}$, $\theta = 10^{-5}$, $\lambda = 500$ nm.

Substituting numerical values, we get:

$$\delta = 81\pi - 800 \frac{\pi}{1\text{m}} \cdot y$$

The maximum corresponds to $\delta = 2\pi n$. at point O there is a minimum.

$$y_1 = \frac{\lambda}{4\alpha} = 1.25\text{mm}$$

$$\Delta y_m = \frac{\lambda}{2\alpha} = 2.5 \text{ mm}$$

Answer:

$$y_1 = 1.25 \cdot 10^{-3} \text{ m} \Delta y_m = \frac{\lambda}{2\alpha} = 2.5 \text{ mm}$$

B6

0.90 pts

Find how the intensity $I(y)$ on the screen depends on the coordinate y if its maximum value is I_0 , substituting here the numerical values from the previous paragraph. Consider the effect of the beam splitter on the intensity of transmitted light.

The field amplitude in beam 1 is $\sqrt{2}$ times less than in the second. Then the intensity is proportional to the square of the amplitude:

$$I(y) = I_0 \frac{1 + 2 + 2\sqrt{2} \cos \delta}{3 + 2\sqrt{2}}$$

$$\delta = 80\pi \left(1 - 10 \frac{y}{1 \text{ m}} \right)$$

Answer:

$$I(y) = I_0 \frac{1 + 2 + 2\sqrt{2} \cos \left[80\pi \left(1 - 10 \frac{y}{1 \text{ m}} \right) \right]}{3 + 2\sqrt{2}}$$

C1

0.40 pts

Express the wave vector $\vec{k}$ of the incident rays in terms of θ_x , θ_y , the modulus of the wave vector $k = 2\pi/\lambda$ and the unit unit vectors of the coordinate axes.

Due to the smallness of the angles θ_x and θ_y :

$$k_z = -k$$

$$k_x = -k\theta_x$$

$$k_y = -k\theta_y$$

Answer:

$$\vec{k} = -k(\theta_x \hat{x} + \theta_y \hat{y} + \hat{z})$$

C2

0.60 pts

Express the phase difference between beams 1'' and 2'' in terms of λ , L_1 , L_2 , L_3 , L_4 , θ_x , θ_y , α , y .

Note that the initial displacement along the x axis of the rays hitting one point on the screen is of the order of $L_3\theta_x$, the difference in the optical paths is correspondingly proportional to θ_x^2 , so the contribution θ_x can be ignored. The answer is similar to the previous part of the problem:

$$\delta_{1''2''} = \frac{2\pi}{\lambda} ((L_1 + L_2)\theta_y + (L_1 + 2L_3 - L_2) - 2\alpha y) + \pi$$

Answer:

$$\delta_{1''2''} = \frac{2\pi}{\lambda} ((L_1 + L_2)\theta_y + (L_1 + 2L_3 - L_2) - 2\alpha y) + \pi$$

Write V as an integral over one variable. It is not necessary to perform integration explicitly. *Hint: the integral of some function $f(\theta_y)$, depending only on θ_y , over the surface of a disk with radius θ_R can be represented as:*

$$I = \int_{-\theta_R}^{\theta_R} d\theta_y \int_{-\sqrt{\theta_R^2 - \theta_y^2}}^{\sqrt{\theta_R^2 - \theta_y^2}} d\theta_x f(\theta_y) = \int_{-\theta_R}^{\theta_R} d\theta_y \cdot 2\sqrt{\theta_R^2 - \theta_y^2} f(\theta_y).$$

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Parts of the star emit independently, so the intensities from them can be added.

$$dI = J d\theta_x d\theta_y \left(3 - 2\sqrt{2} \cos \left(\frac{2\pi}{\lambda} (L_1 + 2L_3 - L_2 - 2\alpha y) + \frac{2\pi}{\lambda} (L_1 + L_2 + d)\theta_y \right) \right)$$

Denote $\varphi_0 = \frac{2\pi}{\lambda} (L_1 + 2L_3 - L_2)$.

$$I = \int_{-\theta_R}^{\theta_R} J \sqrt{\theta_R^2 - \theta_y^2} \left(3 - 2\sqrt{2} \cos \left(-\frac{4\pi}{\lambda} \alpha y + \frac{2\pi}{\lambda} (L_1 + L_2 + d)\theta_y + \varphi_0 \right) \right) d\theta_y$$

Note, that:

$$I_{max} = \int_{-\theta_R}^{\theta_R} J \sqrt{\theta_R^2 - \theta_y^2} \left(3 + 2\sqrt{2} \cos \left(\frac{2\pi}{\lambda} (L_1 + L_2 + d)\theta_y \right) \right) d\theta_y$$

$$I_{min} = \int_{-\theta_R}^{\theta_R} J \sqrt{\theta_R^2 - \theta_y^2} \left(3 - 2\sqrt{2} \cos \left(\frac{2\pi}{\lambda} (L_1 + L_2 + d)\theta_y \right) \right) d\theta_y$$

Then the visibility:

$$V = \frac{2\sqrt{2} \int_{-\theta_R}^{\theta_R} \sqrt{\theta_R^2 - \theta_y^2} \cos \left(\frac{2\pi}{\lambda} (L_1 + L_2 + d)\theta_y \right) d\theta_y}{3 \int_{-\theta_R}^{\theta_R} \sqrt{\theta_R^2 - \theta_y^2} d\theta_y}$$

Denote $\omega =$ Note that the integrable function is even, therefore:

$$V = \frac{8\sqrt{2}}{3\pi} \int_0^1 dw \sqrt{1 - w^2} \cos(\eta w)$$

Answer:

$$V = \frac{8\sqrt{2}}{3\pi} \int_0^1 dw \sqrt{1 - w^2} \cos(\eta w)$$

Find Betelgeuse's angular radius θ_R . Find its radius R if the distance to it is $h = 6.079 \cdot 10^{15}$ km. Hint: The first two zeros of the integral

$$F_n = \int_0^1 (1 - w^2)^n \cos(\eta w) \, dw$$

as a function of η are given in the table below.

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The first visibility resets correspond to η_1 and η_2 for $n = 0.5$. Thus:

$$\eta_2 - \eta_1 = \frac{2\pi}{\lambda}(d_2 - d_1)\theta_R$$

$$\theta_R = \frac{(\eta_2 - \eta_1)\lambda}{2\pi(d_2 - d_1)} = 1.18 \cdot 10^{-7}$$

$$R = \theta_R h = 7.16 \cdot 10^8 \text{ km}$$

Answer:

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